GG-G-00650a(DSA-DM)
September 25, 1970
Interim revision of Fed. Spec.
GG-G-650,
March 11, 1959

INTERIM FEDERAL SPECIFICATION

GRID, RADIOGRAPHIC, STRAIGHT, WAFER-TYPE

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification GG-G-650.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers straight wafer-type radiographic grids, complete with carrying case; for use with an X-Ray apparatus during radiographic examination.

1.2 Classification.

1.2.1 Sizes. Grids shall be for the following film sizes, as specified (see 6.3).

Size I - 10 by 12 inches
Size II - 14 by 17 inches

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 6525

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SPECIFICATIONS

Federal

C-F-206	Felt, Sheet, Wool, Pressed.
NN-P-530	Plywood; Flat-Panel.
TT-E-527	Enamel, Synthetic, Lustreless.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

STANDARDS

Federal

Fed. Std. 102	Preservation, Packaging, and Packing Levels.
Fed. Std. 123	Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. 595	Colors.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Activities otuside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C., 20402.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

(Copies of Military specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Grids. Grids shall be constructed of materials hereinafter specified (see 3.2.1), of the highest quality throughout, and suitable in all respects for the purpose intended.

* 3.1.2 Carrying case.

3.1.2.1 Plywood. Plywood shall be Group B, Exterior Type, Douglas fir conforming to Federal Specification NN-P-530.

3.1.2.2 Felt. Felt shall be an oil-free wool felt conforming to Federal Specification C-F-206.

3.1.2.3 Other materials. Steel shall be used for the channel. The hinge and other hardware shall be fabricated of materials suitable in all respects for the purpose intended.

3.2 Grid.

* 3.2.1 Construction. Size I grid shall be suitable for 10 by 12 inch film, and shall have overall dimensions of 11 by 13 inches. Size II grid shall be suitable for 14 by 17 inch film sizes, and shall have overall dimensions of 15 by 18 inches. A tolerance of ± 1/32 inch shall apply to both overall dimensions. The grids shall have not less than 55 uniformly-spaced lead strips to the inch. Strips shall be parallel to the long dimension. The angulation of the strips shall be nominal focus at 38 inches, so as to provide for a working range of 34 to 44 inches. Minimum transmission through the outer edges shall be not less than 70 percent of the transmission through the center of the grid. The grid ratio shall be 6:1; the width of the lead strips shall be no less than 0.075 inch and the spacing between strips no greater than 0.015 inch. The lead strips shall be sandwiched

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between nonmetallic interspace material that can provide minimal absorption of the X-Ray beam. The lead strips and their dividers shall be firmly cemented between two cover sheets of aluminum, each having a maximum thickness of 0.013 inches. The aluminum covers shall be roll crimped around the entire periphery of the grid. Each corner of the completed grid shall be rounded with a maximum radius of 1/2 inch. The overall thickness of the completed grid shall be 0.130 $\pm$ 0.005 inches. The manufacturer's standard commercial finish shall be acceptable for the exterior cover sheets of the grid. The top cover sheet of the grid shall be suitably identified as "tube side" (see 3.5.1), and shall have an inscribed center line running parallel to the long dimension. Each grid shall be serialized and shall be marked in accordance with 3.5.1. When tested in accordance with 4.4.1, a radiograph of the grid shall show: no marked variations in density, no opaque inclusions, no lead turnover, no doubling nor omission of a lead strip, no pressure marks, nor sectional pattern, (see 3.2.2).

* 3.2.2 Radiograph. One radiograph shall be supplied with each grid and shall include the serial number of the grid. The radiograph shall be taken in accordance with 4.4.1 (see 3.2.1).

* 3.3 Carrying case. Each grid shall be provided with a carrying case in accordance with one of the following types of construction.

3.3.1 Type 1. Type 1 carrying case shall be designed so that the case cover opens at one end to allow the grid to be inserted into a compartment within the case. The case shall be constructed of three layers of 3/8-inch plywood, sound on the exterior sides. The outer layers of the plywood material shall be of one piece and shall form the exterior sides of the case. The inner layer of plywood material(s) shall be constructed to form an inside compartment for housing the grid. The inside compartment of the case shall be completely padded and lined with felt securely cemented in place. Except for the compartment opening, the outer periphery of the case shall be bound on three sides by steel channel 3/64 inches thick that bounds the exterior plywood surfaces for a depth of 1/2-inch on six sides. The plywood layers shall be secured to the steel channel by a series of rivets located 5/16 inches from the edges of the plywood boards. The open end of the case shall have a cutout to allow the fingers of one hand to grasp and remove an inserted grid. A pan shaped hinged cover, fabricated of 3/64 inch thick steel sheet, shall fit over the opening of the case. In the closed position, the cover shall set on top of the exposed ends of the steel channel, and overlap the top exterior surface for a depth of 1/2 inch (not including cutout). The exposed ends of the steel channel shall be recessed from the top of the plywood opening to allow for proper closure. The top inside surface of the cover shall be padded and lined with felt securely cemented in place. The hinge on the cover shall also be attached to one

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end of the steel channel, and allow the cover to be opened to a full 180 degrees. The other end of the cover shall have a catch, and corresponding end of the steel channel shall have a latch to lock the closed cover in place. All exposed metal and plywood edges of the carrying case shall be contoured to eliminate sharp edges and points. There shall be no projections on the large surfaces of the carrying case which interfere with stacking of carrying cases.

3.3.2 Type 2. Type 2 carrying case shall be a hinged cover box design with cushioning inserts. The box shall open bookwise on the long edge of the case. The cover and box shall be pan shaped and shall be constructed of not less than 3/64 inch thick steel sheet. The cover shall fit over the box. The inside of the box shall be lined with 7/8-inch thickness foam cushioning blocks securely cemented in place. The top surface shall be lined with a sheet of felt securely cemented in place. The inside periphery of the box shall be lined with felt strips securely cemented in place, to form a compartment for the housing of the grid. The build-up of foam and felt material around the periphery of the box shall fill the entire depth of the box. One end of the cushioning shall have a cut-out to allow the fingers of one hand to grasp and remove the grid. The inside cover shall be lined with felt securely cemented in place. The felt shall be positioned to cover the cavity in the box and provide additional cushioning for the grid. One long edge of the case shall be supplied with hinges that allow the cover to be opened to a full 180 degrees. The opposite edge of the case shall be supplied with suitable latches and catches to securely lock carrying case when in the closed position. All exposed edges and corners of the case shall be suitably contoured so that there will be no sharp edges and points. There shall be no projections on the large surfaces of the case which interfere with stacking.

3.3.3 Type 3. Type 3 carrying case shall be designed as a hinged holder and lid, so that the lid will open bookwise on the short dimension of the case. The holder shall consist of a frame, and a panel sheet securely spot welded to the inside protruding flange of the frame. The frame shall be formed from closed channels. All corners shall be welded on all contact surfaces. The holder shall be constructed from not less than 1/32 inch thick steel. The frame shall contain slots to accommodate hinges and crossbar locks. The inside panel of the holder shall be lined with a sheet of felt securely cemented in place. The inside panel shall have additional felt cushioning material securely cemented in place, so that there will be no movement of an inserted grid within the closed case; and there shall be sufficient clearance around the frame to allow the grid to be removed after the case is opened. The lid shall be

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fabricated from 1/16 inch steel sheet and shall have the underside lined with a sheet of felt securely cemented in place. The lid shall be hinged to the short side of the frame, and the exterior of the lid shall have attached three flat strip steel spring crossbar locks and a steel D ring - to facilitate opening. The locks shall be attached to the lid by riveting, and shall - when depressed and rotated - engage slots in the holder frame to close and lock the carrying case. The D ring shall be secured by means of a steel attachment bracket secured by rivets, and shall be mounted on end opposite hinges. One end of the frame shall have suitable cutouts to accommodate two floating hinges fabricated of steel, which shall be self adjusting to allow the lid to open to a full 180 degrees, and shall close on top of an inserted grid so that there are no protruding parts above the level of the frame which interfere with stacking. In the closed position, the lid shall not cause any pressure on an inserted grid, and shall fit snugly within the frame without any rattling or motion in any direction. The closed carrying case shall have no protruding parts which interfere with stacking. All exposed edges and corners of the case, slots lock-bars, hinges and bracket shall be suitably contoured so that no sharp edges and points remain.

3.3.4 Drop test. When tested as specified in 4.4.2, the carrying case with grid in place shall be capable of being dropped to a concrete floor without damage to the grid; with the latches on carrying case remaining secure, and cemented felt padding intact. Deformation of short edges (see figure 1) shall not be cause for rejection provided the carrying case opens and closes properly. The tested sample shall not be offered to the Government, as part of the required quantity.

* 3.4 Finish. The carrying case hardware shall be pretreated with a phosphate coating and then shall be dipped or sprayed all over with a synthetic primer. A suitable primer shall be applied to the outside of the carrying case prior to application of the paint. The outside of the case, including hardware shall be finished with two coats of enamel conforming to TT-E-527. The color shall 24087, semigloss, olive drab, as shown in Federal Standard No. 595 (see 6.4).

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3.5 Identification marking.

* 3.5.1 Grid. Each grid shall be marked in a conspicuous location with the stock number, ratings (ratio, lines per inch, focal range), the manufacturer's name or registered trademark, and serial number. The "tube side" shall be permanently inscribed in lower right corner of grid.

* 3.5.2 Carrying case. Each carrying case shall be permanently marked with the stock number, item identification, contract or purchase order number, contractor's name or registered trademark, and the letters "U.S." in suitable size on one large side and on one end of the carrying case.

3.6 Workmanship. Workmanship shall be first class throughout. The grid and carrying case shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the grids and carrying cases may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the Item Identification, the name of the component or material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

* 4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and Table I.

Table I

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major	II	1.0
Total defects (major and minor combined)	II	4.0
For dimensional examination	S-1	2.5

* 4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and Table II.

Table II

Component	Characteristic	Requirement	Test procedure	Inspection** level
Grid	Radiographic Test	3.2.1 & 3.2.2	4.4.1	100%
Grid	Flatness of grid face	3.2.1	4.4.3	S-2
Carrying case with grid	Drop test	3.3.4	4.4.2	***

** Acceptance number shall be zero (0) regardless of lot size.
***One unit per lot.

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4.3 Examination. The grid and carrying case shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects as showing in Table III.

Table III

Categories		Defects**
Major A (Unit of product-Grid)		
	101	Grid covers not free of cracks, tears and deformed surfaces.
	102	Surfaces of grid not flat or not free of warping.
	103	Grid not free of rattling which indicates that lead strips and dividers are not securely cemented between aluminum covers.
	104	Grid not free of sharp edges or points.
	105	Marking on grid not present, complete or legible.
Major B (Unit of product-carrying case)		
	151	Carrying case not capable of being opened and closed properly.
	152	Grid not capable of being inserted or removed from case.
	153	Felt padding not securely cemented to inside of carrying case.
	154	Latches or spring lock do not properly close secure carrying case.
	155	Latches or spring lock do not open on carrying case.
	156	Carrying case not free of projections which interfere with stacking.
	157	Carrying case not free of sharp edges or points.
	158	Steel channel on Type 1 carrying case not securely bound around sides of plywood carrying case.
	159	Lid or case not free of warping.

** Examinations shall not be restricted to the possible classified defects listed above.

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Table III (cont'd.)

Categories	Defects**
Minor A (Unit of product-Grid)	
201	Grid not free of dirt, grease or other foreign matter.
202	Finish on grid not uniform or not free of scratches or marks.
203	Center line on grid not present, legible, or continuous.
Minor B (Unit of product-carrying case)	
251	Felt lining not free of dirt, grease or other foreign matter.
252	Finish on carrying case not uniform or not free of scratches or marks.
253	Carrying case not free of dirt, grease or other foreign matter.
254	Surface of carrying case not free of indentations, burrs, cracks or tears.
255	Identification marking not present, legible or complete.
256	Hinges not firmly secured to carrying case.

* 4.3.2 Dimensional examination. The grids and carrying cases shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

* 4.4 Tests. Tests shall be conducted to determine compliance with all requirements contained in this specification.

* 4.4.1 Radiographic test. The radiograph shall be made at a target-to-film distance of 38 inches with central X-ray beam vertical to the plane of the grid and centered on the grid, and a maximum applied voltage of 80 KVP.

* 4.4.2 Drop test. The carrying case with grid in place shall be subjected to the following test:

(a) (4) - Four 18 inch drops to a concrete floor (see Fig 1(a)), once on each short edge of the carrying case, with the carrying case positioned like a diamond so that the line made up of two corners and one short edge as per figure will be parallel to the floor with the center of gravity directly over this line.

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(b) (2) - Two 36 inch drops to a concrete floor, with the case landing flat, once on each large side (see Fig 1(b)).

Radiographic test as per 4.4.1 shall be taken after drop test and compared with original radiograph.

4.4.3 Flatness of grid face. The test for flatness shall be conducted on both faces of the grid. The grid shall rest face down on a flat test surface, and deviations from flatness of the opposite surface shall be determined by passing feeler gages between that surface and a straightedge placed at various angles on the surface being tested.

5. PREPARATION FOR DELIVERY

5.1.1 Unit of issue. One grid, as specified, constitutes one unit of issue.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with Fed Std No. 123.

5.3.2 Military

5.3.2.1 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The grid is for use with mobile, portable, and field radiographic units.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.1 and 5.2).
(c) Size required (see 6.3).

6.3 This specification covers the following items listed in the Federal Supply Catalog:

Stock Number	Item Identification
6525-603-1250	Grid, Radiographic, 10 by 12 inches, Straight Wafer Type, Size I
6525-603-1310	Grid, Radiographic, 14 by 17 inches, Straight Wafer Type, Size II.

* 6.4 Availability of color chips. The individual 3 by 5 inch chip of the color referenced in 3.4 is available from General Services Administration, Business Service Center, Region 3, Seventh and D Streets, S. W. Washington 25 D. C.. Price 5 cents.

6.5 This specificaton does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

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6.6 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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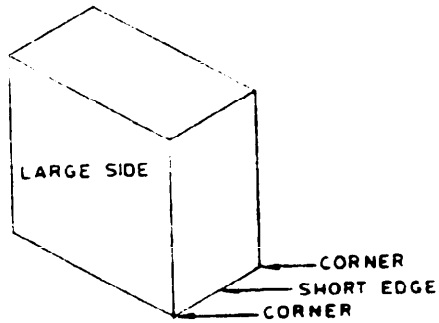


FIGURE 1

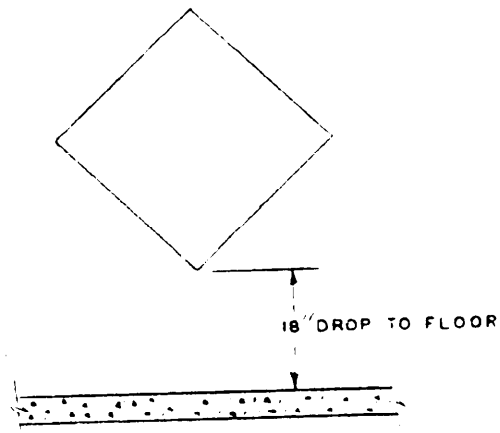


FIGURE 1 (a)

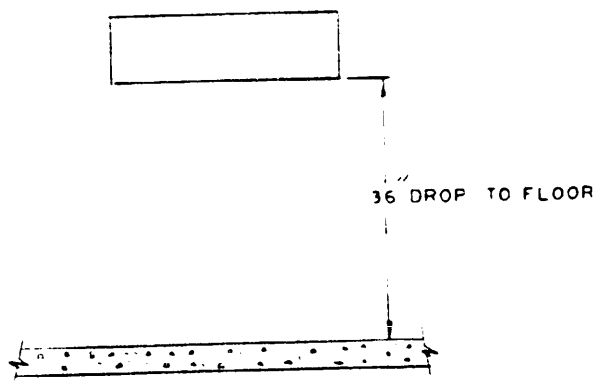


FIGURE 1 (b)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION: GG-G-00650a(DSA-DM) FSN:

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426 1 JAN 66 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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